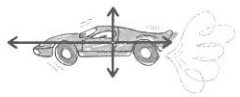
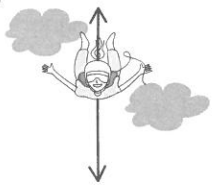
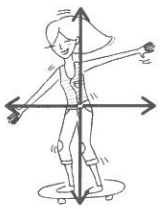
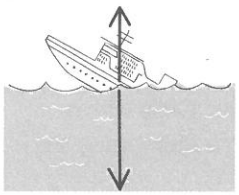
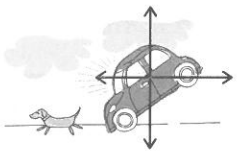
7.1 What can forces do?

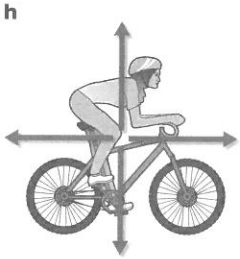
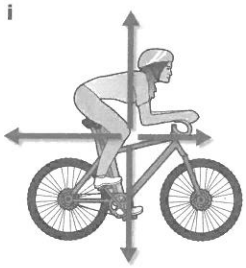
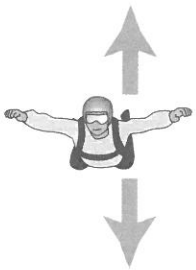
Science understanding

Visual/Spatial Verbal/Linguistic

LB 7.1 p.142

Forces can be pushes, pulls or twists. A force can be represented by an arrow pointing in the direction it is acting. The bigger the arrow, the bigger the force. Many forces can act on an object at the same time. If these forces are not balanced, the motion of the object will change. **Analyse** the forces acting on each object below and complete the table.

Situation	Are the forces balanced?	If not balanced, state the direction the unbalanced force acts.	Describe any changes to this object's motion.
a 	☐ Yes ☑ No	☐ Up ☐ Down ☑ Left ☐ Right	The car moves to the left.
b 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
c 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
d 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
e 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
f 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	

Situation	Are the forces balanced?	If not balanced, state the direction the unbalanced force acts.	Describe any changes to this object's motion.
g 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
h 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
i 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
j 	☐ Yes ☐ No	☐ Up ☐ Down	
k 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	
l 	☐ Yes ☐ No	☐ Up ☐ Down ☐ Left ☐ Right	

7.2 Car safety features

Science as a human endeavour

Verbal/Linguistic

Refer to the Science as a Human Endeavour on pages 249–50 of your student book to answer the following questions.

1 **(a)** **Describe** the aims of ANCAP (Australasian New Car Assessment Program).

_____

_____

(b) **Explain** how cars are rated.

_____

_____

2 Use Newton's First Law of Motion to **explain** why passengers travelling in a car that hits a solid object are thrown forward.

_____

_____

3 **Propose** reasons why you should *not* place heavy objects on the ledge behind the back passenger seat in a car.

_____

_____

4 In terms of Newton's First Law, **explain** why dogs travelling in a car should be kept in a restraint.

_____

_____

5 **Explain** the difference between active and passive safety features.

_____

_____

6 **Classify** the following safety features as active or passive: airbags, reversing sensors, ABS (anti-lock brakes), traction control, seat belts, crumple zones, brake assist, ESC (electronic stability control).

Active features					
Passive features					

7 Modern cars are designed with crumple zones that will crumple if the car is in an accident. **Explain** why these crumple zones are beneficial.

crumple zone (n) part of a car that is designed to be squashed in an accident

8 Active safety features are currently being developed. For example:

- frontal pre-collision safety warning systems: a radar detects when a collision is going to happen and alerts the driver or applies the brakes
- lane department warning systems: alert drivers when they move out of a lane
- adaptive cruise control: a radar detects how fast the car in front is travelling and adjusts the car's speed to keep a safe distance away
- eye tracking technology: detects when a driver is tired and sounds an alarm
- side impact protection: radars and cameras predict whether a side impact is going to happen, and automatically slide a metal rod across this side of the car.

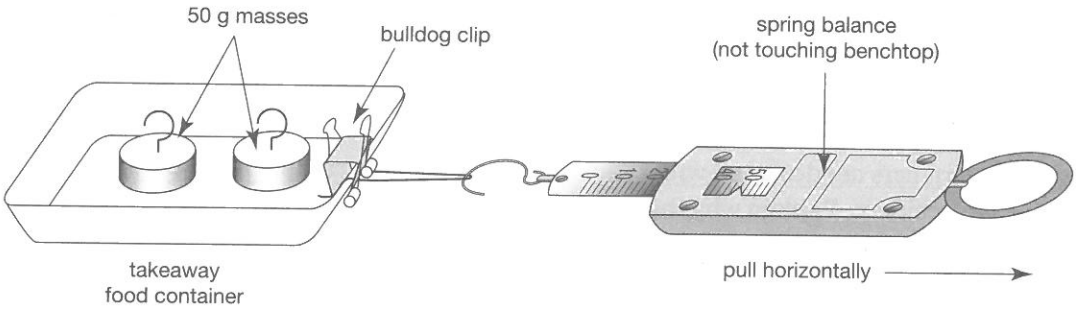
(a) **Discuss** which of these ideas you think offers the most benefit to passenger safety.

(b) **Propose** your own idea for an active car safety feature.

7.3 Measuring friction

Science inquiry

Logical/Mathematical Visual/Spatial



Chang and Amalia completed an experiment to see whether adding mass to a tray affected the amount of friction when they tried to slide it along a bench top. The tray and bulldog clip were measured to have a mass of 100 grams. The students pulled the tray and clip carefully and measured the reading on the spring balance while the tray was sliding. Their results for the experiment are shown below.

Total mass of tray and weights (g)	Sliding friction (N)			Average sliding friction (N)
	Trial 1	Trial 2	Trial 3	
100	0.3	0.5	0.4	
150	0.8	0.8	0.8	
200	1.4	1.5	1.0	
250	1.6	1.9	1.9	
300	2.3	2.2	2.4	
350	2.5	2.5	3.1	
400	3.1	3.4	3.1	
450	3.6	3.7	3.5	

1 **Propose** why Chang and Amalia measured three readings of sliding friction for each mass tested.

2 Add the forces measured as Trial 1, Trial 2 and Trial 3 and divide your result by 3 to **calculate** the average sliding friction (N) in each case. Complete the last column of the table. Round each average to one decimal place.

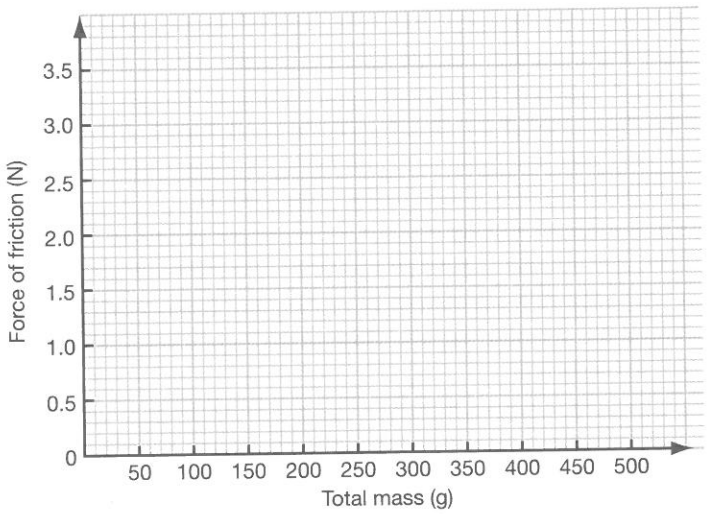
3 Chang and Amalia suspected that they made an error when measuring one result. **Identify** the likely error by circling this result in the table.

4 **Propose** how the students may have recorded an incorrect reading.

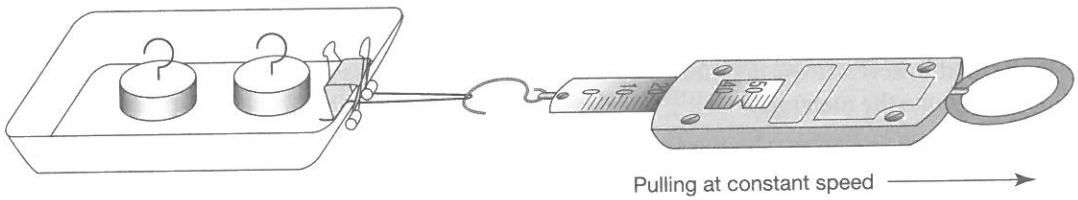
_____

_____

5 **Construct** a graph of your results on the axes provided.



6 (a) **Draw** arrows on the diagram below to show four forces acting on the tray while it is being pulled at a constant speed.



(b) **State** whether the forces acting on the tray are balanced or unbalanced at this time.

_____

LB 7.3 P.143

7 **Propose** a conclusion for Chang and Amalia's experiment.

Chang and Amelia concluded that the _____ the mass, the _____ the friction between the tray and the bench top. Therefore, the _____ the force required to slide the tray at a constant speed.

HINT

Choose from these words:

greater	lower
lighter	heavier

8 **Propose** a real-world example where you have noticed a link between friction and mass.

_____

_____

7.4 Fast suit controversy

Science as a human endeavour

Verbal/Linguistic

Fast suits out but not their records

1 ROME is known as the Eternal City and after a controversial ruling on swimming costumes by the sport's governing body, it may well be that the records set there in the pool over the next week at this year's world championships will last forever.

2 In a decision that has sparked widespread anger among swimmers – including past champions and future contenders – FINA has ruled that the latest high-tech, full-body swimsuits will be banned from next year but that records set using the super-fast suits made of buoyant polyurethane will be allowed to remain in the history books.

3 The decision, announced in Italy on Friday, was supposed to draw a line under a bitter row that has consumed the sport in recent years, although it may ultimately cause more controversy.

4 Olympic legend Grant Hackett labelled FINA's ruling an 'absolute debacle', saying: 'It's terrible that they've allowed the situation to get to this point where it is overshadowing what should be the biggest event in the sport.'

5 The debate over the technology of costumes – and whether body suits give swimmers who wear them an unfair edge – has been raging since 1992 when Speedo launched the world's first 'fast suit'. But the issue has intensified since last year when the same maker launched the NASA-designed LZR costume made in part from polyurethane, which allegedly traps pockets of air and aids buoyancy. The effect was immediate, with 108 world records falling last year alone.

6 But that technology appears to have been bettered in 2009 as Adidas, Arena and Jaked introduced suits covered in polyurethane. A further 27 world records have fallen this year.

7 Some members of the Australian team competing in Italy will be wearing the controversial suits, including Olympic champion Stephanie Rice and butterfly swimmer Andrew Lauterstein.

8 Hackett believes FINA's decision to permit the latest suits at the world championships will undermine the meeting.

9 'They should have been banned and it should have been done a long time before now,' he said. 'You won't be able to compare these records to those that have gone before ... These suits are giving a bigger advantage than the greatest performance-enhancing drugs.

10 'It will also be a blow to those athletes who do break world records ... if they're using these suits, people will always ask how much was down to their ability and dedication and how much down to technology.'

11 But FINA executive director Cornel Marculescu said records and those set this week would remain because they reflected performances by athletes in the circumstances of their time. 'There is no way we are going to reclaim any times. They are existing world records.'

12 Under rule changes to be announced tomorrow, suits will revert to the kind worn a decade ago, before swimmers such as Hackett and Ian Thorpe made full-body suits a popular choice.

13 FINA is expected to ban the use of polyurethane and decree that male suits will only be able to cover the area between the waist and the knee, while female suits will not be able to extend beyond the shoulder.

14 Australia's 100-metre freestyle world record holder, Eamon Sullivan, is also critical of the decision to allow the new suits in competition and fears it could undermine the sport for the next 20 years.

15 'When these suits get banned, it's going to take a very long time before athletes get to a point where they can swim these record times without that technology,' Sullivan said. 'Who knows how long we'll have to wait for another world record? Ten, 15, maybe 20 years? That's going to take a lot of excitement from the sport.'

16 Former double Olympic champion Kieren Perkins said FINA should have banned full-length suits when they were first developed. 'When they introduced the first suits I and many others were very vocal that this was a slippery slope that we were about to go down.'

The Age, 26 July 2009

controversial ruling (n) a decision that causes argument

draw a line under (v) mark the end of

row (n) an argument

consumed (v) taken up time and energy

absolute debacle (n) a complete mess

undermine (v) to weaken

reclaim (v) to take back

revert (v) to change back

slippery slope (n) the start of something that will become a problem

High-tech swimsuits made from polyurethane began to appear in the early 1990s. These swimsuits were controversial because some people believed they gave the swimmer an unfair advantage in the pool.

1 **Recall** the number of world records that were broken in 2008. (paragraph 5)

_____

2 **Propose** how the all-polyurethane suits may have given swimmers an unfair advantage. (paragraphs 2 and 5)

_____

_____

3 Although the all-polyurethane suits are now banned, the records set by swimmers who were wearing them have been allowed to remain. **Discuss** why some people are not happy about this ruling. (paragraphs 9 and 10)

_____

_____

_____

_____

4 Some of the swimsuits also had an ultra-low profile zip instead of a regular zip. **Propose** how this would be an advantage for a swimmer.

ultra-low profile (adj) very flat

_____

_____

5 **List** one argument for and one argument against keeping the world records set by swimmers wearing the high-tech suits. (paragraphs 9, 10 and 11)

For: _____

_____

_____

Against: _____

_____

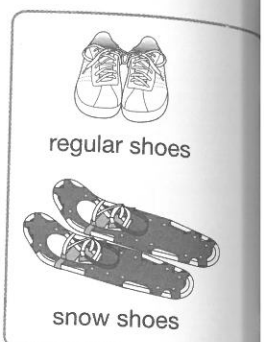
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7.5 Feeling the pressure

Science understanding, Science inquiry

Logical/Mathematical Visual/Spatial Bodily/Kinesthetic

It is hard to walk on soft snow in regular shoes without sinking, but you can walk on soft snow while wearing a pair of wide snow shoes. This is because your weight force is spread over a wider area. Polar bears have very large feet for this reason. Pressure, P, is the amount of force per unit area. It is measured in a unit called the pascal (Pa) and can be calculated:



$$P = \frac{F}{A}$$ where F is force (N) and A is area (m^2).

1 **Calculate** the pressure you exert on the ground when standing in bare feet.

(a) On the next page, trace the outline of your left foot on the grid paper

(b) Count the number of 1 cm squares contained within the outline of your foot. (Estimate and add part squares to the number of whole squares.)

Area of left foot = _____ cm^2

To convert this value into square metres, divide it by 10 000 (or move the decimal point four places to the left).

Area of left foot = _____ m^2

Assuming your feet are the same size, multiply this result by 2.

Area of both feet = _____ m^2

(c) **Calculate** the size of your weight force by multiplying your mass (in kilograms) by 10 (approximate gravitational field strength).

My weight force = _____ N

(d) **Calculate** the pressure you exert on the floor by dividing your weight (in newtons, N) by the area of both feet (in m^2). (Note: Round your answer off to one decimal place.)

$$P = \frac{F}{A} = \frac{\text{weight force (N)}}{\text{area of both feet } (m^2)} = _____ = _____ \text{ Pa}$$

2 **Predict** whether the pressure you exert on the ground would increase or decrease when you are standing:

(a) in a pair of slippers three sizes larger than your own _____

(b) on your tip toes to reach for a biscuit jar _____

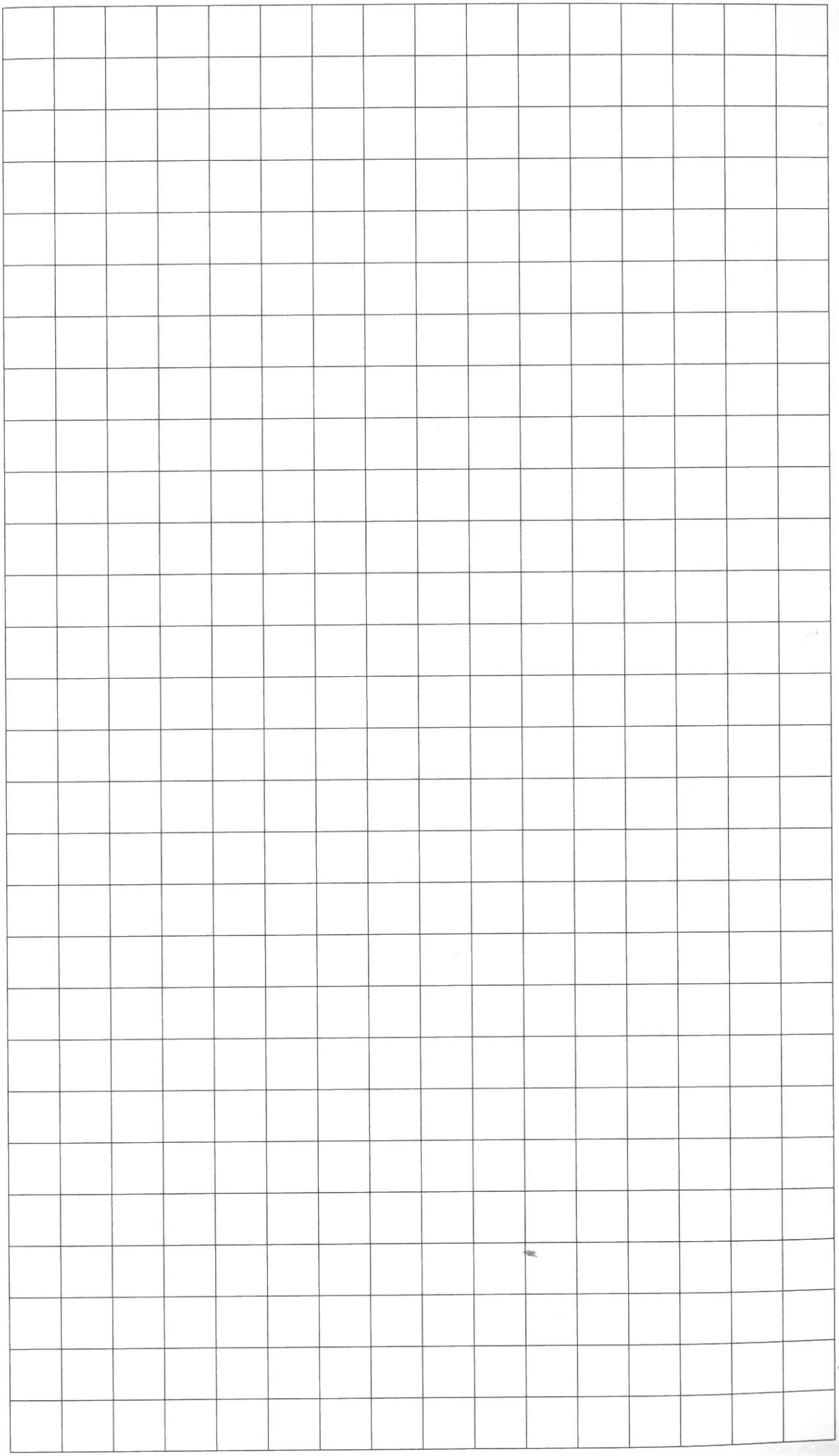
(c) in a pair of spiked football boots. _____

3 **Propose** why a drawing pin can be pushed easily into a corkboard, but a coin of the same mass and pushed with the same force cannot.

_____

4 **Analyse** why high-heeled shoes, particularly stilettos, can damage a wooden floor.

_____



7.6 Buoyancy

Science understanding

Verbal/Linguistic Visual/Spatial

If you push an inflated balloon under water, it will come back up. You can feel a force pushing the balloon upwards as you try to force it down. This is because the water exerts an upwards force on the balloon. This force is called *buoyancy*. The Greek mathematician Archimedes discovered that the size of this force is equal to the weight of water that has been displaced (pushed aside) by the object.

inflated (adj) full of air

exert (v) to use force or power

Whether something sinks or floats depends upon its *density*. Density is a measure of the mass that is packed into a volume.

Look at the steel ball in Figure 7.6.1. This ball has quite a small volume, so it displaces a small weight of the water. The upwards buoyancy force is less than the gravity pulling the ball down, so it sinks. In contrast, the hull of a ship has a larger volume and so displaces a large weight of water. This produces a large buoyancy force (greater than the weight of the ship), and so the ship floats. If the ship was made from solid steel, its weight force would be greater and it would sink. In other words, if the ship has a lower density (mass per volume) than the water, it will float. If it has a higher density, it will sink.

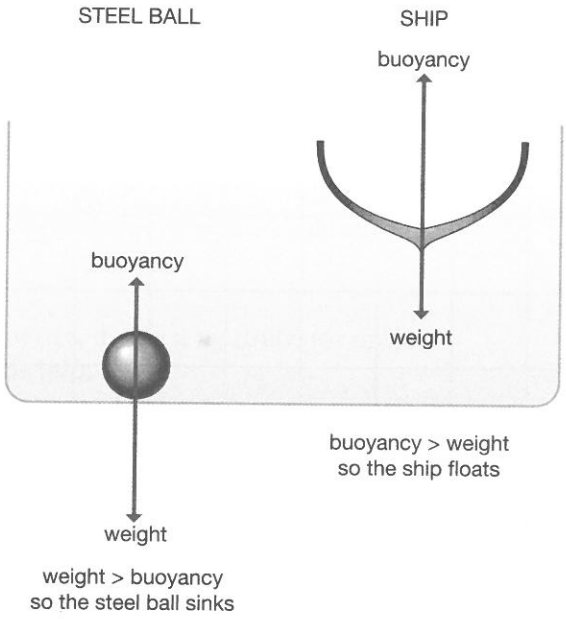


Figure 7.6.1 The relative sizes of the upwards buoyancy force and downwards weight force result in the steel ball sinking and the ship staying afloat.

Sea water has a greater density than fresh water. This means that sea water gives you more buoyancy than the fresh water of a lake or swimming pool. The Dead Sea, located on the Israel–Jordan–Palestine border, is one of the saltiest seas on Earth. As a result, it exerts such a large buoyancy force that it is impossible for a person to sink!

Ships that carry cargo around the world travel through waters of different saltiness. Some ports may be less salty because of the inflow of fresh water from rivers. For this reason, all large ships have markings on the side. These markings show the safe level that the ship can be pushed into the water with its cargo to account for buoyancy in different waters. This is shown in Figure 7.6.2.

cargo (n) the goods carried by a ship, train or plane

account for (v) to make allowance for

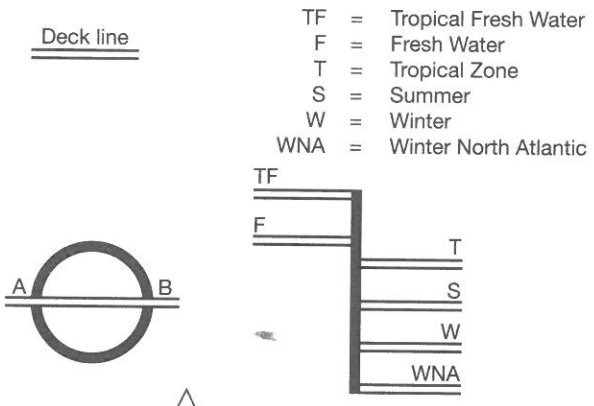
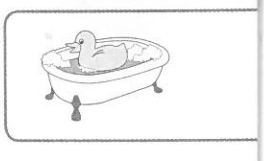


Figure 7.6.2 International Load Lines for ships



1 **Name** the force that acts upwards on a rubber duck floating in a bath.

_____

2 **Identify** the correct answer.

(a) If the buoyancy on an object is greater than its weight, then the object will float/sink.

(b) If the buoyancy on an object is less than its weight, then the object will float/sink.

3 **State** whether:

(a) sea water has a greater or lesser density than fresh water

_____

(b) the buoyancy force of salt water is greater or less than that of fresh water.

_____

4 **Assess** whether the following objects would float or sink in water.

(a) a leaf _____

(b) a marble _____

(c) a sheet of paper _____

(d) an apple _____

(e) a football _____

(f) a padlock _____

(g) a cork _____

5 A ship can be described as a thin shell with lots of air inside. **Explain** how this helps the ship to float.

_____

_____

6 **(a)** **Design** an experiment to **compare** the buoyancy of objects in water with the buoyancy of objects in olive oil. **List** the steps.

_____

_____

_____

(b) **Name** the variables in the experiment.

_____

_____

7.7 Maglev trains

Science understanding, Science as a human endeavour

Verbal/Linguistic

A maglev (short for 'magnetic levitation') train has no wheels, but it travels at up to 500 km/h! It floats, or levitates, a few centimetres above its track as it moves its passengers quietly, smoothly and very quickly. Some maglevs levitate because of magnetic repulsion. This force exists between powerful magnets (called superconducting magnets) beneath the train and on top of its track (called a guideway). The magnets in the guideway shown in Figure 7.7.1 are electromagnets. Their poles continually reverse in order to keep repelling the train and pushing it forwards. Other types of maglevs use a section of the train that wraps around the guideway and propels the train forward by magnetic attraction.

pole (n) the part of a magnet where the force is strongest

repel (v) to make something move away by force

Maglev trains use less electricity than conventional trains, can turn in smaller circles and climb steeper hills. They are less noisy, faster and have less wear and tear on parts. However, the installation of a maglev system is expensive, because of the technology inside its guideway, which can be very long.

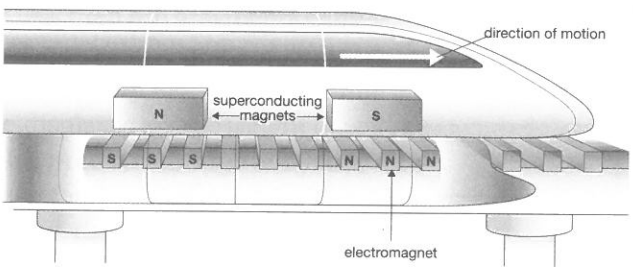


Figure 7.7.1

This maglev is pushed upwards by magnetic repulsion.

1 **Explain** why these trains are called maglevs.

2 **Explain** why a maglev train only needs to overcome friction from the air as it travels.

3 A maglev train has no engine. **Describe** two ways that a maglev can be made to move forwards.

4 **List** three advantages of maglev trains compared to electric trains.

5 **State** the main disadvantage of the maglev system.

7.8 Static electricity

Science understanding

Visual/Spatial

The list on the right forms part of the triboelectric series. This lists materials in order of how easily they lose electrons. Materials above cotton tend to lose electrons easily, and may become positively charged. Materials below cotton gain electrons easily and so become negatively charged.

Air
Human hands
Asbestos
Glass
Human hair
Nylon
Wool
Fur
Lead
Silk
Aluminium
Paper
Cotton
Steel
Wood
Hard rubber
Nickel, copper
Silver
Gold
Acetate
Polyester
Styrofoam
Polyethene
Vinyl
Teflon

1 **State** whether air is likely to lose or gain negative charges.

_____

2 You rub a plastic acetate sheet against your hair, and your hair is attracted to it.

(a) **State** which has become positively charged.

_____

(b) **State** which has become negatively charged.

_____

(c) **Explain** why your hair is attracted to the acetate.

_____

_____

3 When styrofoam is charged with a piece of silk, **identify** which material loses electrons.

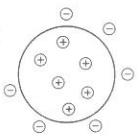
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4 On a warm and dry day, the fur of Josef's pet mouse is seen to be attracted to Josef's vinyl shoes. **Explain** what is happening in this situation.

_____

_____

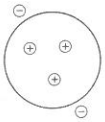
5 Use these diagrams to **state** the number of positive and negative charges in each atom, and **classify** each as having a positive, negative or no overall charge.

(a) 

Positive charges: _____

Negative charges: _____

Overall charge: _____

(b) 

Positive charges: _____

Negative charges: _____

Overall charge: _____

7.9 Literacy review

Science understanding

Verbal/Linguistic

Recall your knowledge of forces by correctly inserting words from the list below to complete the statements below. Use each word once.

attract	change	direction	weight	field
friction	gravitational	inertia	mass	negatively
newton	pole	positively	pull	repel
shape	slow	spring	stop	surface

1 A force can be a push, a _____ or a twist.

2 A force can _____ an object's motion by making an object start moving, _____ moving, change _____, change _____, speed up or _____ down.

3 A force is measured using a _____ balance by using a unit called the _____.

4 The tendency for an object to keep its current state of motion is called its _____.

5 When pushing a refrigerator across the floor, you need to overcome the force of _____. This force exists whenever one _____ slides over another.

6 A force of _____ attraction exists between your body and the Earth.

7 Your _____ is a downwards force due to gravity.

8 _____ is the amount of matter in a substance.

9 A magnet has a north and a south _____.

10 Like magnetic poles will _____ while unlike magnetic poles will _____.

11 An object that loses electrons becomes _____ charged.

12 An object that gains electrons becomes _____ charged.

13 The region around an object where a non-contact force is experienced is called a _____.